FF-W-84a
April 17 1967
SUPERSEDING
Fed. Spec. FF-W-84
March 13, 1951

FEDERAL SPECIFICATION

WASHERS, LOCK (SPRING)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.** This specification covers a helical spring type washer for use with nuts, bolts, screws, and studs to prevent them from loosening.

1.2 **Classification.** Washers covered by this specification shall be of the following classes and styles, as specified (see 6.2)·

Class A—Carbon steel
- Style 1—Light.
- Style 2—Medium.
- Style 3—Heavy.
- Style 4—Extra heavy.

Class B—Corrosion-resistant steel (magnetic).
- Style 1—Light.
- Style 2—Medium
- Style 3—Heavy.

Class C—Corrosion-resistant steel (nonmagnetic).
- Style 1—Light.
- Style 2—Medium.

Class D—Aluminum-zinc alloy (7075).
- Style 2—Medium.

Class E—Phosphor bronze or tin brass.
- Style 1—Light
- Style 2—Medium.

2. APPLICABLE DOCUMENTS

2.1 **Specifications and standards.** The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein

Federal Specifications:

QQ-A-225/9—Aluminum Alloy Bar, Rod, Wire, and Special Shapes; Rolled, Drawn, or Cold Finished 7075.

QQ-B-750—Bronze, Phosphor; Bar, Plate, Rod, Sheet, Strip, Flat Wire, and Structural and Special Shaped Sections.

QQ-N-286—Nickel-Copper-Aluminum Alloy, Wrought (K-Monel).

QQ-P-35—Passivation Treatments for Austenitic, Ferritic, and Martensitic Corrosion-Resisting Steel (Fastening Devices).

QQ-P-416—Plating, Cadmium (Electrodeposited).

QQ-Z-325—Zinc Coating, (Electrodeposited), Requirements For.

Federal Standards:

Fed Std No 66—Steel: Chemical Composition and Hardenability

Fed Std. No. 123—Marking For Domestic Shipment (Civilian Agencies).

Fed. Std. No. 595—Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S Government Printing Office. Washington, D.C, 20402

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas

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(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-A-8625—Anodic Coatings, For Aluminum and Aluminum Alloys
MIL-T-10727—Tin Plating, Electrodeposited or Hot-Dipped, For Ferrous and Nonferrous Metals
MIL-P-16232—Phosphate Coatings, Heavy, Manganese or Zinc Base.
MIL-I-17214—Indicator, Permeability, Low-Mu
MIL-H-3982—Hardware (Fasteners and Related Items), Packaging and Packing for Shipment and Storage of

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129—Marking For Shipment and Storage.
MS-35338—Washer, Lock, Split, Helical, Medium Series
MS-35340—Washer, Lock, Split, Helical, Extra Heavy Series

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

United States of America Standards Institute (USASI) Publication:

B27.1—Lock Washers.

(Application for copies should be addressed to the United States of America Standards Institute, 10 E. 40th St., New York, N. Y., 10016.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) Standards

A 153—Zinc Coating (Hot Dip) on Iron and Steel Hardware

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, Pa., 19103.)

Copper Development Association (CDA)

CDA 425

(Application for copies should be addressed to the Copper Development Association, 405 Lexington Ave., New York, N. Y., 10017.)

(Technical society for technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 **Material.**

3.1.1 *Class A.* Unless otherwise specified (see 6.2), class A spring lockwashers made of carbon steel shall be compositions 1060 to 1080 in conformance to the requirements of Fed. Std. No. 66. Class A washers shall be properly quenched from above the critical temperature and tempered to a hardness of 45 to 53 Rockwell C. Depending upon the cross section, there may be used the Rockwell C, D, or A scale, the superficial Rockwell N scale, or the Vickers hardness tester, and the readings converted to Rockwell C. The washers shall not have a thickness of decarburized surface of more than 0.002 inch or 1-1/2 percent of the sectional thickness, whichever is the greater.

3.1.2 *Class B.* Unless otherwise specified (see 6.2), class B spring lockwashers made of corrosion-resisting steel shall be of composition 420 in conformance with the requirements of Fed. Std. No. 66. After coiling, the class B washer shall be heat treated and quenched in air or oil, as indicated by the proportions of the washer section, and tempered at a temperature not to exceed 800°F. to a hardness of 43 to 53 Rockwell C.

3.1.3 *Class C.* Unless otherwise specified (see 6.2), class C spring lockwashers made

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of corrosion-resisting steel shall be of the 300 series in conformance with the requirements of Fed Std No 66 As the result of cold working during fabrication, the washer shall have a hardness of 35 to 43 Rockwell C The class C spring lockwashers shall have a magnetic permeability of 2 0 maximum (air=1 0) for a field strength of A= 200 Oersteds using a magnetic indicator per MIL-I-17214

3 1 4 *Class D.* Unless otherwise specified (see 6.2), class D spring lockwashers shall be made of 7075 aluminum alloy in conformance to the requirements of QQ-A-225 '9 The washers shall receive a solution and precipitation heat treatment resulting in a hardness of 75 to 97 Rockwell B, or a minimum tensile strength of 77,000 p s i Tensile test specimens of keystone wire heat treated with the washers they represent, shall be furnished when specified in the invitation for bids.

3.1.5 *Class E.* Unless otherwise specified (see 6.2), class E spring lockwashers shall be made of phosphor bronze in conformance to the requirements of QQ-B-750, composition A, or tin brass, CDA alloy No 425 As the result of cold working during fabrication, class E washers shall have a minimum hardness of 90 Rockwell B

3.1.6 *Class G* Unless otherwise specified (see 6 2), class G spring lockwashers shall be made of nickel-copper-aluminum alloy in conformance to the requirements of QQ-N-286, class A The washer shall be fabricated from cold-worked material and subsequently age hardened to a hardness of 33 to 40 Rockwell C or a minimum tensile strength of 155,000 p s i

3.1 7 *Brass-bronze washers.* Brass-bronze washers shall be dyed a deep yellow in accordance with Fed Std No 595, color No. 23538

3.2 Design.

3.2 1 *Washer section* The finished washer cross section shall be slightly trapezoidal in shape with the internal and external circumferential edges rounded sufficiently to avoid heat checks The thickness at the inner periphery shall be greater than the outer periphery by an amount varying from a minimum of 0 0005 to a maximum of 0 001 inch per 1 64 inch of section width The minimum thickness specified in USA B27.1 represents the nominal mean thickness of the trapezoid In the formula, the increase in thickness from the outer periphery to inner periphery shall be from 0.032 W (minimum) to 0 064 W (maximum)

3.2.2 *Dimensions* When the dimensions are not otherwise specified, styles 1, 2, 3 and 4 spring lockwashers shall conform to those shown in USA B27 1 When specified (see 6.2), the dimensions shall conform to the applicable dash number of MS 35338 and MS 35340

3.2.3 *Tolerances.* Unless otherwise specified (see 6 2), the tolerances on all dimensions shall be as shown in USA B27.1, or applicable MS standards

3 2 4 *Free height.* The washers shall be coiled so that the free height is approximately twice and not less than 1-2/3 times the mean thickness of the washer section The severed ends shall not overlap or be positioned so that the washers will interlink when tested as specified in 4 5.3

3.3 Protective finishes. Unless otherwise specified (see 6.2), spring lockwashers shall be furnished uncoated When cadmium, zinc, phosphate, tinned, anodic, or passivation films are specified, they shall conform to the following paragraphs (see 6.2)

3.3.1 *Cadmium plating* When specified (see 6 2), cadmium plating shall be in accordance with QQ-P-416, type II, class 3

3.3.2 *Zinc coating (electrodeposited)* When specified (see 6.2), zinc coating by electrodepositing process shall be in accordance with QQ-Z-325, type II, class 3

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3.3.3 *Zinc coating (hot-dipped).* When specified (see 6.2), zinc coating by hot-dipping process shall be in accordance with ASTM A 153.

3.3.4 *Phosphate coating* When specified (see 6.2), phosphate coating shall be in accordance with MIL-P-16232, type z.

3.3 5 *Anodic coating.* When specified (see 6.2), anodic coating shall be in accordance with MIL-A-8625

3.3.6 *Tin coating* When specified (see 6.2), tin coating shall be in accordance with MIL-T-10727

3.3 7 *Passivation* Passivation treatments shall conform to the requirements of QQ-P-35

3.4 **Hydrogen embrittlement.** Unless otherwise specified, electroplated washers shall be baked for 4 hours at 400°F immediately after plating.

3.5 **Cleaners.** The cleaners used shall not etch the material and allow subsequent cracking of the washers. Acid cleaners shall not be used Scale shall be removed by mechanical methods

3.6 **Workmanship.** The flat faces of washers and the inner and outer peripheries shall be smooth and free from knurling, serrations, die marks, deep scratches, etc., although slight feed roll marks are permissible. The washers shall have rounded edges and be free from burrs, rust, pit marks, and other defects which may affect their appearance or serviceability The ends of the gaps shall not present cutting edges at the contact surfaces The workmanship shall be in accordance with high-grade commercial practice covering this class of work.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Lot.** For the purpose of sampling, a lot shall consist of not more than 100,000 wasners in sizes up to 7/8 inch and not more than 20,000 washers in sizes above 7/8 inch manufactured of the same material, type, and size produced under essentially the same conditions and presented at any one time for inspection in connection with the same contract or order

4.3 **Sampling.**

4.3 1 *Sampling for examination.* For purposes of examination, as specified in 4 4.1, samples shall be selected from each lot in accordance with inspection level II of MIL-STD-105 and acceptable quality level (AQL) of 4.0.

4.3.2 *Sampling for acceptance tests.* For purposes of testing, as specified in 4 4 1, samples shall be selected from each lot in accordance with inspection level S-1 of MIL-STD-105 and of AQL 2.5

4.3.3 *Sampling for test of protective finish* Sampling for test of protective finish shall be as specified in the applicable specification of 3.3

4 3.4 *Sampling for inspection of packaging and packing.* Sampling for examination of section 5 requirements shall be as specified in MIL-H-3982.

4.4 **Inspection.**

4.4.1 *Visual and dimensional examination* Each sample shall be visually and dimensionally examined for conformance with 3.2 If the number of rejected samples exceeds the acceptable number, the lot represented by the sample shall be rejected

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4.4.2 *Acceptance testing.* Any sample that fails to conform to the acceptance limits of 4.3.1 shall be cause for rejection of the lot represented

4.5 Tests.

4.5.1 *Interlinking.* The samples shall be subdivided into groups of two each. An attempt shall be made with each group to manipulate them so that interlocking or interlinking occurs. The washers shall not be bent nor deformed in performing this test. Any interlinking or interlocking of these washers shall be considered a defect.

4.5.2 *Free height test* The free height of the washer shall be measured to the nearest 0.001 inch (see 3.2.4). The washer shall be subjected to a compression, twice pressed flat, released, and the free height again measured. The free height after the second compression shall be at least two-thirds of the original free height (or five-sixths) when so specified in the invitation for bid.

4.5.3 *Twist test.* The twist test shall be conducted in accordance with USA B27.1.

4.5.4 *Pressure test.* The washers shall be clamped between hardened flat steel plates. They shall be held flat for one minute and then released and inspected The free height of each washer shall be at least two-thirds (or five-sixths, if so specified in the invitation for bids) of the original free height and shall be unimpaired for the lot to be accepted. If all are unimpaired, they shall again be clamped flat, held for 16 hours (or longer), and then released and inspected for the third time Failure of any sample washer to withstand any of these tests shall be cause for rejection of the lot which it represents.

4.5.5 *Hardness test.* All sample washers shall be tested for hardness in accordance with the requirements of 3.1 Before making the hardness test on class A washers, both surfaces shall be ground parallel and decarburization shall be removed.

4.5.6 *Decarburization of class A washers.* The affected depth of decarburization of carbon steel washers, as specified in 3.1, shall be determined by microscopic examination at 100 magnifications.

4.5.7 *Chemical analysis.* Samples selected in accordance with 4.3 shall be analyzed to determine compliance with 3.1.1 through 3.1.6, inclusive, as applicable Failure of any sample to conform to the requirements of this specification shall be cause for rejection of the entire lot represented.

4.5.8 *Examinations and test of preparation for delivery.* The preservation, packaging, packing, and marking for military procurement, shall be examined or tested in accordance with MIL-H-3982.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing The washers shall be preserved and packaged level A or C and packed level A, B, or C of MIL-H-3982. The level of protection shall be as specified in the contract or order (see 6.2).

5.2 Marking.

5.2.1 *Civil agencies.* In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed Std No. 123.

5.2.2 *Military agencies.* In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129

6. NOTES

6.1 Intended use. The spring lockwashers covered by this specification are intended for automotive and general bolt, nut, and screw applications They have the dual function of (1) spring take-up devices to compensate for developed looseness and the loss of tension between component parts of an assembly, and (2) hardened thrust bearings

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to facilitate assembly and disassembly of fastenings by decreasing the frictional resistance between the bolted surface and the bearing face of the bolt or screwhead or nut

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(*a*) Title number and date of this specification
(*b*) Class, style of washer required (see 1.2)
(*c*) Type of protective coating, if required (see 3.3.1 through 3 3 7)
(*d*) Military standard part number, if applicable (see 3.2.2)
(*e*) Tolerances (see 3.2 3).
(*f*) Dimensions (see 3.2.2)
(*g*) Material (see 3.1.1 through 3 1 6).
(*h*) Selection of applicable levels of packaging and packing required (see 5.1)

6.2.1 *For civil agency procurements* When level B packaging is required for civil agency procurements, the level A requirements in 5 1 shall apply

6.3 For military procurements. Styles 1 and 3 are not recommended for new design, since they will not be available after a transition period (see MS 16214, MS 35337, and MS 35339) For new procurements see MS 35338 and MS 35340

6 4 **Supersession data** Class A style 5 and class B style 5, washer and machine screw assemblies have been deleted from this document and will be covered by FF-S-001321 for Screw, Assembled Washer Until this document is printed, it is suggested that USA B18.13 for Screw and Washer Assemblies-Sems be used for procurement

CUSTODIANS:
Army—WC
Navy—WP
Air Force—69

Review activities:
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Navy—WP SH SA
Air Force—69 85

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User activities.
Army—MO
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5 PROBLEM AREAS

a Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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